Notes: Hachem & Song (JPE, 2021)

Liquidity Rules and Credit Booms

Contents

1 Big picture

- **Question:** Why can *tighter* liquidity regulation (e.g., a higher reserve requirement / stricter loan-to-deposit cap) coincide with a *credit boom*?
- **Core mechanism:** When (i) a large bank has *market power* in the interbank market and (ii) smaller banks can *arbitrage* the liquidity rule via shadow banking, tighter regulation induces smaller banks to expand shadow banking and offer higher "incentives" to savers. This shifts deposits/funding *away* from the large, liquid bank toward smaller, less-liquid banks. Because aggregate liquidity must cover the worst aggregate-withdrawal state, this reallocation can *lower* required aggregate liquidity and *raise* aggregate lending.
- **Empirical application:** China. The paper documents interbank market power (repo rates move with big-bank funding announcements) and argues that tightening/strict enforcement of liquidity rules contributed to the post-2008 rise of shadow banking and the aggregate credit-to-savings ratio.

2 Model 1: Benchmark economy (no liquidity regulation)

2.1 Timeline and balance sheet

- Three dates $t = 0, 1, 2$.
- Two "banks" in the reduced-form model:
  - j: representative **interbank price taker** (stands for a continuum of small banks).
  - k: an **interbank price setter** (a large bank with market power in the interbank market).
- Total initial funding is normalized to 1. Bank k has funding share $x_k^0 \in (0, 1)$; bank j has $1 - x_k^0$.
- At $t = 0$, bank $i \in \{j, k\}$ chooses a **liquidity ratio** $\lambda_i \in [0, 1]$:
  - liquid reserves: λ_i (per unit of funding),
  - illiquid loans/projects: $1 - \lambda_i$ (per unit of funding).

2.2 Loan technology

Banks invest the illiquid share in a concave return technology $g(\cdot)$.

Assumption 1 (Properties of g)**.** *Let $g : [0, 1] \to \mathbb{R}_+$ satisfy: $g(0) = 0$; $g(x)/x > 1$ for $x \in (0, 1]$; $g'(x) > 0$, $g'(1) > 1$; $g''(x) < 0$; and $g'''(x) \geq 0$.*

Interpretation: lending is profitable on average but has decreasing marginal returns.

2.3 Liquidity shocks and the interbank market

At $t = 1$, withdrawals arrive. There are two non-crisis states $s \in \{A, B\}$ with probabilities $\pi_A, \pi_B \in (0, 1)$ and a crisis state C with probability $\varepsilon \equiv 1 - \pi_A - \pi_B$ (used later in the welfare discussion). In state s, a fraction $\theta_i^s \in (0, 1)$ of bank i's funding is withdrawn at $t = 1$.

In the benchmark analysis, focus on the *non-crisis* states. Let

$$\tilde{\pi} \equiv \frac{\pi_A}{\pi_A + \pi_B}, \qquad \mathbb{E}(r) \equiv \tilde{\pi} r_A + (1 - \tilde{\pi}) r_B.$$

Assume the shocks are arranged so that j tends to need liquidity in A while k tends to need liquidity in B:

$$\theta_j^A > \max\{\theta_k^A, \theta_j^B\}, \qquad \theta_k^B > \max\{\theta_k^A, \theta_j^B\},$$

and expected shocks are equalized across banks:

$$\tilde{\pi}\theta_j^A + (1 - \tilde{\pi})\theta_j^B = \tilde{\pi}\theta_k^A + (1 - \tilde{\pi})\theta_k^B.$$

Aggregate feasibility. In each non-crisis state s, total reserves must cover total withdrawals. With funding shares $(x_k^0, 1 - x_k^0)$,

$$(\lambda_k - \theta_k^s)x_k^0 + (\lambda_j - \theta_j^s)(1 - x_k^0) \geq 0, \qquad s \in \{A, B\}. \tag{1}$$

2.4 Interbank price taker (j): portfolio choice given interbank rates

Let $r_s \geq 0$ denote the net interbank rate in state s (so lending 1 at $t = 1$ pays back $1 + r_s$ at $t = 2$).

Per unit of funding, bank j's ex-post payoff in state s is

$$\Upsilon_j^s(\lambda_j; r_s) = g(1 - \lambda_j) + \lambda_j - 1 + r_s(\lambda_j - \theta_j^s). \tag{2}$$

It chooses λ_j to maximize $\tilde{\pi}\Upsilon_j^A + (1 - \tilde{\pi})\Upsilon_j^B$. The first-order condition (interior) is

$$g'(1 - \lambda_j) = 1 + \mathbb{E}(r). \tag{3}$$

Intuition: marginal return to making one more unit illiquid (reducing λ_j) equals the marginal cost of losing one unit of liquidity plus the expected interbank cost of being short.

$$\lambda_j = \cdots \tag{3}$$

2.5 Interbank price setter (k): choosing (λ_k, r_A, r_B)

Bank k internalizes that r_A and r_B affect $\mathbb{E}(r)$ and hence, through (3), the *price takers'* choice of λ_j.

Taking the small banks' λ_j as given, bank k's expected payoff is

$$\Upsilon_k \equiv \big[g(1 - \lambda_k) + \lambda_k - 1\big]x_k^0 + \Big[\tilde{\pi} r_A(\theta_j^A - \lambda_j) + (1 - \tilde{\pi}) r_B(\theta_j^B - \lambda_j)\Big](1 - x_k^0). \tag{4}$$

The interbank component is positive in the state in which k lends and negative in the state in which k borrows.

Which feasibility constraint binds? Let s' denote the state with the largest aggregate withdrawals:

$$s' \in \arg\max_{s\in\{A,B\}} \Big(\theta_j^s(1-x_k^0) + \theta_k^s x_k^0\Big).$$

An important simplification is that k never optimally holds slack aggregate liquidity in *both* states.

Lemma 1 (Aggregate feasibility binds in the worst state)**.** *In any optimum,* (**??**) *binds in at least one state. In particular, if s' denotes the highest-withdrawal state, any optimal choice satisfies*

$$(\lambda_k - \theta_k^{s'})x_k^0 + (\lambda_j - \theta_j^{s'})(1-x_k^0) = 0. \tag{5}$$

Interest-rate bounds. The paper imposes $r_s \in [0, \bar{r}_s]$ to capture frictions and limits in interbank pricing.

2.6 Equilibrium definition and benchmark results

Definition 1 (Unregulated equilibrium)**.** *An unregulated equilibrium is a tuple $(\lambda_j^*, \lambda_k^*, r_A^*, r_B^*)$ such that:*

1. *λ_j^* solves* (3) *given $\mathbb{E}(r^*)$;*
2. *$(\lambda_k^*, r_A^*, r_B^*)$ maximize* (4) *subject to* (5) *and the bounds on r_s.*

Proposition 1 (Cross-sectional liquidity)**.** *In any equilibrium with $\lambda_j^* \in (\theta_j^B, \theta_j^A)$ and $r_B^* > 0$, the interbank price setter is more liquid:*

$$\lambda_k^* > \lambda_j^*.$$

Intuition: the price setter internalizes the effect of its own liquidity on its *borrowing costs* when it is short (state B), and therefore self-insures more than the price takers.

Proposition 2 (Existence (and why $r_B^* > 0$ may require a cap on r_A))**.** *There exists a unique equilibrium with $\lambda_j^* \in (\theta_j^B, \theta_j^A)$ and $r_B^* > 0$ when θ_j^A is sufficiently high and the lending-rate cap $\bar{r}_A$ is sufficiently low.*

Interpretation: If k could always raise r_A without limit, it would use r_A to manipulate $\mathbb{E}(r)$ and set r_B as low as possible. A low $\bar{r}_A$ ensures that k eventually "runs out of room" in state A, so that in equilibrium it also chooses $r_B > 0$.

3 Model 2: Liquidity regulation and shadow banking

3.1 Liquidity floor and regulatory arbitrage

Regulation imposes a liquidity floor $\alpha \in (0,1)$, interpreted as a minimum reserve ratio or, equivalently, a maximum loan-to-deposit ratio $(1-\alpha)$.

A bank can engage in regulatory arbitrage by pushing some funding off balance sheet through shadow banking. Let $\xi_i \geq 0$ be an *incentive* (a per-unit-of-funding cost) and let $h(\xi_i) \in [0,1)$ be the fraction of funding that becomes "shadow" and escapes the liquidity rule.

Assumption 2 (Shadow banking technology)**.** *$h(0) = 0$, $h'(\xi) > 0$, $h''(\xi) \leq 0$, and $h'''(\xi) \geq 0$.*

If the regulator applies the liquidity floor to *reported* (on-balance-sheet) funding $(1 - h(\xi_i))$, the *effective* constraint on the *true* liquidity ratio is

$$\lambda_i \geq \alpha\big(1 - h(\xi_i)\big). \tag{6}$$

Thus, for fixed α, higher ξ_i (more shadow banking) relaxes the required true liquidity.

3.2 Funding reallocation (deposit competition)

Funding can move across banks depending on incentives. The reduced-form share equation is

$$x_k = x_k^0 + \delta_1\big(\xi_k - \bar{\xi}_j\big), \tag{7}$$

where $\bar{\xi}_j$ is the average incentive offered by the continuum of price takers and $\delta_1 > 0$ measures how sensitive savers are to incentive differences. In a symmetric equilibrium among price takers, $\xi_j = \bar{\xi}_j$.

3.3 Price taker problem with regulation

For bank j, the choice variables are (λ_j, ξ_j) with multipliers μ_j on the liquidity constraint and ρ_j on $\xi_j \geq 0$.

The paper's Kuhn–Tucker conditions can be summarized as:

$$g'(1 - \lambda_j) = 1 + \mathbb{E}(r) + \mu_j, \tag{8}$$

$$\rho_j = 1 - \alpha\mu_j h'(\xi_j), \tag{9}$$

$$\mu_j\big[\lambda_j - \alpha(1 - h(\xi_j))\big] = 0, \quad \mu_j \geq 0, \quad \lambda_j \geq \alpha(1 - h(\xi_j)), \tag{10}$$

$$\rho_j\,\xi_j = 0, \quad \rho_j \geq 0, \quad \xi_j \geq 0. \tag{11}$$

When does shadow banking emerge? Define two cutoff expected rates:

$$\underline{R}(\alpha) \equiv g'(1 - \alpha) - 1 - \frac{1}{\alpha h'(0)}, \qquad \overline{R}(\alpha) \equiv g'(1 - \alpha) - 1.$$

Lemma 2 (Binding constraint and shadow banking thresholds)**.** *Bank j is constrained by regulation ($\mu_j > 0$) iff $\mathbb{E}(r) < \overline{R}(\alpha)$. Shadow banking emerges ($\xi_j > 0$) iff $\mathbb{E}(r) < \underline{R}(\alpha)$. Moreover, ξ_j is decreasing in $\mathbb{E}(r)$.*

Intuition: if interbank liquidity is cheap (low $\mathbb{E}(r)$), the bank wants to be illiquid, so it hits the floor and then starts using shadow banking to reduce its *true* liquidity requirement.

3.4 Regulated equilibrium and the credit-boom result

Definition 2 (Regulated equilibrium)**.** *A regulated equilibrium is $(\lambda_j, \lambda_k, \xi_j, \xi_k, r_A, r_B, x_k)$ such that: (i) price takers solve (8)–(11) taking $(r_A, r_B, \xi_k, \bar{\xi}_j)$ as given, (ii) x_k satisfies (7), (iii) the price setter chooses $(\lambda_k, \xi_k, r_A, r_B)$ to maximize its payoff subject to feasibility and anticipating the price takers' best responses.*

Two key intermediate results:

- Price takers can start shadow banking when α tightens above their unregulated choice.
- For small enough deposit-competition intensity δ_1, the price setter prefers *not* to use shadow banking (it is not constrained, and shadow banking is just a costly way to defend market share).

Lemma 3 (No shadow banking by the price setter (locally))**.** *There exists $\bar{\delta}_1 > 0$ such that for $\delta_1 \in (0, \bar{\delta}_1)$, in a neighborhood of $\alpha = \lambda_j^*$, bank k optimally sets $\xi_k = 0$.*

Aggregate liquidity and aggregate credit. Define aggregate liquidity

$$\mathrm{LIQ} \equiv \lambda_j(1 - x_k) + \lambda_k x_k,$$

so aggregate credit is $1 - \mathrm{LIQ}$.

Because feasibility binds in the highest-withdrawal state s', aggregate liquidity equals required aggregate withdrawals in that state:

$$\mathrm{LIQ} = \Theta^{s'}(x_k), \qquad \Theta^{s'}(x_k) \equiv \theta_j^{s'} + (\theta_k^{s'} - \theta_j^{s'})x_k.$$

Proposition 3 (Credit boom from tighter liquidity regulation)**.** *Fix $\delta_1 \in (0, \bar{\delta}_1)$. When the liquidity floor α is perturbed upward above λ_j^*, aggregate credit rises (equivalently,* LIQ *falls)* if and only if *the binding feasibility state is $s' = B$.*

Intuition: when the tight aggregate-liquidity state is the one where the large bank has the bigger withdrawal need ($\theta_k^B > \theta_j^B$), shifting deposits away from the large bank reduces the system's required liquidity buffer.

Decomposing the change in aggregate liquidity. Differentiate LIQ:

$$\frac{d\,\mathrm{LIQ}}{d\alpha} = (1 - x_k)\frac{d\lambda_j}{d\alpha} + x_k\frac{d\lambda_k}{d\alpha} + (\lambda_k - \lambda_j)\frac{dx_k}{d\alpha}.$$

The last term is the **reallocation channel**. Since $\lambda_k > \lambda_j$ and tighter regulation induces $dx_k/d\alpha < 0$ (funding shifts toward the high-ξ price takers), the reallocation term pushes LIQ *down* and credit *up*. Whether this translates into a *net* credit boom depends on which state binds (captured by s').

3.5 Additional implications and identification of market power

Proposition 4 (Additional results and the role of interbank market power)**.** *With interbank market power, the following can simultaneously occur as α rises above λ_j^*: (i) a credit boom, (ii) convergence in* on-balance-sheet *liquidity ratios, and (iii)* $\mathrm{corr}(\xi_j, \mathbb{E}(r)) > 0$*. These three features cannot all hold in an economy where all banks are interbank price takers (even allowing heterogeneity in g).*

Interpretation: with market power, the large bank can *raise* interbank rates to deter regulatory arbitrage and defend its funding share, generating positive comovement between shadow-banking incentives and interbank rates.

4 Welfare, regulation, and policy extensions

4.1 Why regulate liquidity at all? (crisis state)

Introduce the crisis state C (probability ε) where everyone withdraws: $\theta_j^C = \theta_k^C = 1$. Unmet withdrawals create a social loss $\kappa(\cdot)$ (increasing, convex).

The planner chooses liquidity subject to feasibility in A and B:

$$\max_{\lambda_j,\lambda_k} g(1-\lambda_j)(1-x_k^0) + g(1-\lambda_k)x_k^0 - \varepsilon\,\kappa\big(1-\lambda_k x_k^0 - \lambda_j(1-x_k^0)\big)\,.$$

Proposition 5 (Inefficiency in decentralized equilibrium)**.** *Suppose $s' = B$. Aggregate liquidity in the decentralized equilibrium is inefficiently low if $\varepsilon > 0$ and*

$$\kappa'\big(1-\Theta^B(x_k^0)\big) > \frac{g'\big(1-\Theta^B(x_k^0)\big)}{\varepsilon}. \tag{12}$$

Otherwise aggregate liquidity is efficient. Regardless, the cross-bank distribution *of liquidity is inefficient.*

4.2 Optimal liquidity floor without shadow banking

Proposition 6 (Implementing the planner when $h \equiv 0$)**.** *If there is* no *shadow banking technology ($h(\xi) \equiv 0$), then a liquidity floor $\alpha = \lambda^*$ can implement the planner's allocation in a competitive equilibrium (under a mild condition ensuring the floor is binding in the right way).*

4.3 With shadow banking and market power

Proposition 7 (Shadow banking and market power can break implementability)**.** *When shadow banking is feasible, a competitive equilibrium can still be implemented with some $\alpha > \lambda^*$. But with interbank market power, there may be no such α, and increasing α above λ_j^* can be welfare-reducing by triggering a credit boom and lowering aggregate liquidity.*

4.4 Bank capital

The paper shows the mechanism survives when banks can use equity, and when capital requirements apply to on-balance-sheet risky assets. (Main takeaway for me: the key is still "measured" liquidity rules $\Rightarrow$ arbitrage $\Rightarrow$ funding reallocation interacting with interbank market power.)

4.5 Central bank liquidity injections

Let the central bank inject liquidity in state s according to a policy rule $\psi(r_s - r^*)$ with $\psi > 0$, where r^* is a target rate. The aggregate feasibility condition becomes

$$(\lambda_j - \theta_j^s)(1-x_k) + (\lambda_k - \theta_k^s)x_k + \psi(r_s - r^*) \geq 0. \tag{13}$$

Proposition 8 (Central bank can implement constrained efficiency)**.** *As $\psi \to \infty$, setting $r^* = g'(1-\lambda^*) - 1$ implements the planner's allocation (intuitively, the central bank pins down interbank rates, eliminating the price setter's market power wedge).*

5 Quantitative application: China

5.1 Evidence for interbank market power: repo and NCD announcements

The paper uses announcements of large-bank NCD issuance as a proxy for large banks' liquidity demand. It estimates:

$$\text{repo}_t = \beta_0 + \beta_1 \text{NCD}_{B4,t-1} + \beta_2 \text{repo}_{t-1} + \beta_3 \text{repo}_{t-2} + \beta_4 \text{RRR}_t + \varepsilon_t. \tag{14}$$

Table 1: Interbank repo rate regressions (China)

	(1)	(2)
$L1.$ repo	0.827***	0.826***
	(0.0377)	(0.0377)
$L2.$ repo	0.0693*	0.0618
	(0.0383)	(0.0384)
RRR	0.209***	0.202***
	(0.0713)	(0.0711)
$L.\,\text{NCD}_{B4}$	0.0394***	
	(0.0120)	
$L.\,\text{NCD}_{ABC}$		0.0227*
		(0.0118)
$L.\,\text{NCD}_{BOC}$		−0.0210
		(0.0161)
$L.\,\text{NCD}_{CCB}$		0.0414***
		(0.0122)
$L.\,\text{NCD}_{ICBC}$		−0.0135
		(0.0475)
Observations	748	748
R^2	0.842	0.844

Notes: Dependent variable is the interbank repo rate (daily). $L1$ and $L2$ denote one- and two-day lags. RRR is the required reserve ratio. NCD variables are dummies for NCD issuance announcements (lagged one day). Standard errors in parentheses. * $p < 0.1$, *** $p < 0.01$.

Interpretation (Table 1).

- A Big-Four NCD announcement predicts a higher repo rate the next day.
- This is consistent with large banks being *systemic* liquidity players that move interbank rates (a reduced-form signature of market power / strategic price setting).

5.2 Calibration and policy experiment

The quantitative model is a disciplined version of the theory (two states, concave lending returns, shadow banking technology, and an external liquidity sector).

A key experiment: tighten the liquidity floor from $\alpha = 0.145$ (around 2007) to $\alpha = 0.25$ (interpretable as the 75% loan-to-deposit cap).

Table 2: Calibration results

	Baseline ($\alpha = 0.145$)		Tight ($\alpha = 0.25$)	
	Model	Data (2007)	Model	Data (2014)
Average interbank rate, $\mathbb{E}(r)$ (%)	0.2	0.2	0.6	1.3
Price-setter LDR, $1 - \lambda_k$	0.62	0.62	0.71	0.70
Credit-to-savings ratio (%)	72.5	72.5	78.7	82.5

Notes: The credit-to-savings ratio is $1 - \lambda_j - (\lambda_k - \lambda_j)x_k$ in the model (aggregate credit as a share of aggregate savings).

Interpretation (Table 2).

- The model matches 2007 targets by construction, then predicts 2014 outcomes under a tighter liquidity floor.
- Tightening increases interbank rates and raises the price setter's LDR (the big bank becomes less liquid), consistent with the model's reallocation/strategic-response channel.
- The model generates a substantial rise in aggregate credit relative to savings (a "credit boom"), accounting for a sizeable fraction of the observed change.

5.3 Time-series comovement: interbank rates and shadow banking incentives

The paper measures ξ_j, ξ_k using 3-month WMP yields (small banks vs Big Four). Table 3 reports the data correlations.

Table 3: Correlations in the data (monthly, 2008–2014)

Correlation	Estimate (s.e.)
$\text{corr}(\mathbb{E}(r), \xi_j)$	0.456 (0.077)
$\text{corr}(\mathbb{E}(r), \xi_k)$	0.329 (0.095)
$\text{corr}(\mathbb{E}(r), \xi_j - \xi_k)$	0.259 (0.093)
$\text{corr}(\xi_j, \xi_k)$	0.736 (0.052)

Notes: $\mathbb{E}(r)$ is a volume-weighted average interbank repo rate. ξ_j and ξ_k are 3-month WMP yields offered by joint-stock commercial banks and the Big Four, respectively.

Interpretation (Table 3).

- All correlations are positive: when interbank liquidity is more expensive, WMP yields (shadow-banking incentives) are higher.
- This is consistent with the model's market-power implication that k may raise interbank rates while j expands shadow banking (positive comovement).

5.4 Structural estimation of shocks

To fit the time-series correlations, the paper allows for three shocks:

- a liquidity-regulation shock: α varies,
- a lending-profitability (credit-demand) shock: z varies,
- an external-liquidity shock: L varies.

For example, the regulatory shock is modeled as

$$\alpha = \bar{\alpha} + \varepsilon_\alpha. \tag{15}$$

Table 4: Estimation results (matching correlation moments)

	Only σ_α	Only σ_z	Only σ_L	$\sigma_\alpha, \sigma_z, \sigma_L$	Data
Panel A. Parameter values					
σ_α	0.0131			0.0396	
	(0.0001)			(0.0030)	
σ_z		0.0037		0.0002	
		(0.0004)		(0.0000)	
σ_L			0.0059	0.0014	
			(0.0002)	(0.0007)	
SSR	1.052	6.382	3.886	0.021	
Panel B. Pairwise correlations					
$\text{corr}(\mathbb{E}(r), \xi_j)$	0.955	−0.999	−1.000	0.371	0.456
$\text{corr}(\mathbb{E}(r), \xi_k)$	0.784	0.861	−0.960	0.352	0.329
$\text{corr}(\mathbb{E}(r), \xi_j - \xi_k)$	0.994	−0.959	0.568	0.296	0.259
$\text{corr}(\xi_j, \xi_k)$	0.931	−0.849	0.960	0.846	0.736

Notes: The table reports simulated moments for models with different shock structures, and the data moments from Table 3. "SSR" is the sum of squared residuals in the moment-matching exercise.

Interpretation (Table 4).

- **Only regulation shocks** get the *signs* right but make correlations too close to 1 in magnitude.
- **Only loan-demand shocks** deliver counterfactual negative correlations between $\mathbb{E}(r)$ and small-bank WMP yields.
- **Only external-liquidity shocks** also generate counterfactual signs for some moments.
- **A mixture of shocks** (column 4) produces correlations closer to the data with a much lower SSR.

6 Short summary

- Liquidity regulation can backfire in the presence of regulatory arbitrage *and* market power in the interbank market.
- The model gives a clean empirical signature of market power: positive comovement between interbank rates and shadow-banking incentives alongside a credit boom and convergence of measured liquidity ratios.
- Policy implication: liquidity floors alone may not deliver more “true” liquidity; central bank liquidity facilities that pin down interbank rates (or broader regulation of shadow banking) matter.

7 Conclusion

7.1 Summary

This paper concludes that **liquidity regulation can backfire and trigger a credit boom** when two ingredients are present:

1. **interbank market power** (a large “price setter” bank that can strategically affect interbank rates), and
2. **regulatory arbitrage via shadow banking** (smaller “price taker” banks can circumvent the rule by shifting funding off balance sheet at a cost).

Under these conditions, tightening a liquidity floor (or stricter enforcement of an LDR-type rule) induces price-taking banks to expand shadow banking and offer higher incentives to attract funding. This **reallocates deposits/funding away from the more liquid price setter toward less liquid price takers**. Because **aggregate liquidity is pinned down by the state with the largest aggregate withdrawals**, this reallocation can **reduce required aggregate liquidity buffers**, thereby **increasing aggregate credit**. In short: *the liquidity rule reshapes who holds deposits, and that reshaping can lower systemwide liquidity demand.*

Beyond the core reallocation channel, the paper concludes that **endogenous portfolio responses** (changes in banks' chosen liquidity ratios) can meaningfully amplify or attenuate the boom. Finally, the paper's China application suggests that the mechanism is empirically relevant: tightening/stricter enforcement of liquidity rules can help explain the rise of shadow banking and a higher credit-to-savings ratio.

7.2 Economic intuition: the mechanism as one continuous chain of logic

The mechanism can be summarized as the following sequence:

1. **Liquidity rules bind asymmetrically.** In equilibrium, price takers are endogenously less liquid than the price setter. A tighter liquidity floor therefore binds first (and most strongly) for price takers.

2. **Shadow banking converts measured compliance into true noncompliance.** When the rule applies to on-balance-sheet funding, a price taker can pay an incentive cost (e.g., higher WMP yields) to move part of funding off balance sheet, reducing the amount of true liquidity it must hold to satisfy the regulation.

3. **Deposit competition shifts funding toward the arbitraging banks.** The same incentives that facilitate off-balance-sheet activity also attract deposits. Thus, tightening the rule shifts funding away from the price setter and toward the price takers.

4. **Aggregate liquidity is determined by the worst withdrawal state.** Systemwide reserves must cover withdrawals in the state with the largest aggregate withdrawals. If the worst state is the one in which the price setter has the larger withdrawal need, reallocating funding *away* from the price setter *reduces* systemwide required liquidity.

5. **Less required liquidity means more lending.** Since aggregate credit is the complement of aggregate liquidity (credit $= 1 - \text{LIQ}$ in the model), a fall in required aggregate liquidity mechanically increases aggregate lending: a **credit boom**.

7.3 What the China evidence implies (how the paper supports the mechanism)

The paper's empirical content supports the model through three main facts:

1. **Interbank market power is plausible.** Table 1 shows that interbank repo rates move with big-bank NCD announcement activity (consistent with large-bank liquidity actions moving market rates rather than being purely price-taking).

2. **Tightening the liquidity floor can quantitatively raise credit relative to savings.** Table 2 shows that raising the liquidity floor from $\alpha = 0.145$ (2007) to $\alpha = 0.25$ (2014 enforcement) increases the model's credit-to-savings ratio from 72.5 to 78.7, compared to an observed rise to 82.5. A convenient benchmark is that the model explains about $6.2/10.0 \approx 62\%$ of the observed increase.

3. **Rates and shadow-banking incentives comove positively.** Table 3 documents positive correlations between interbank rates and small-bank vs big-bank WMP yields. Table 4 shows that matching these correlations requires a mixture of shocks (regulation, lending profitability, and external liquidity), which is consistent with the idea that regulation is a key driver but not the only force in the data.

7.4 Policy implications (what the paper teaches about regulation design)

The paper's policy lesson is not that liquidity rules are useless, but that **partial regulation invites circumvention**:

- A liquidity floor that applies primarily to *on-balance-sheet* activity can push intermediation into **shadow banking**, while also reshuffling funding toward banks that exploit the loophole. This can **raise aggregate credit and lower true system liquidity**, even though measured ratios may appear compliant.

- Because interbank market power shapes the equilibrium interbank rate and thereby the incentives to arbitrage, **market structure in short-term funding markets** is central to how liquidity regulation transmits.

- A central bank liquidity facility (modeled as effectively pinning interbank rates via strong injections when rates deviate from target) can neutralize the market-power distortion and help implement a constrained-efficient allocation in the model.

7.5 Takeaway

With interbank market power and shadow-banking arbitrage, tightening liquidity rules can reallocate deposits away from liquid big banks toward less-liquid small banks, lowering required aggregate liquidity in the worst withdrawal state and thereby generating an unintended credit boom.